

Spatial outliers as a pattern determinant for explaining heterogeneity

Model: Second-Dimension Outlier-Driven Heterogeneity (SOH)

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or:

```
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Overview

This repository contains the data, code, and figure-generation scripts used in the study "Spatial outliers as a pattern determinant for explaining heterogeneity".

The study proposes the **second-dimension outlier-driven heterogeneity (SOH)** model, which incorporates spatial outlier patterns (SOPs) into spatial heterogeneity analysis. SOH derives multi-scale

SOPs from second-dimension outlier information and evaluates their explanatory power through a stratification–detection workflow based on GeoDetector-based PD (Power of Determinant) metrics.

The project integrates Google Earth Engine (GEE), R-based spatial analysis, GeoDetector-based PD calculation, and GIS processing to construct SOPs and assess their contribution to explaining spatial heterogeneity in Australian barley production.

0. Repository structure

```

figureshare/
├── 01_original data_from GEE/
│   ├── 1.1_gee_env_grid_2023_2024.js
│   ├── 1.2_gee_soil_extract.js
│   └── shp
│       ├── East_Large_polygon_pro.shp
│       └── SA2_East_grid.shp
├── 02_original barley_from ABS/
│   ├── 2.1_ABS_Original_SA2_Barley_2023_2024.csv
│   ├── 2.2_SA2_Barley_2023_24.csv
│   ├── 2.3_data_aggregated.R
│   └── Statistical Area Level 2 (SA2) ASGS Ed 2016 Digital Boundaries in
ESRI Shapefile Format/
│   └── SA2_2016_AUST.shp
├── 03_core codes/
│   ├── 3.1_soh_generate_sops.R
│   ├── 3.2_calculate_pd.R
│   ├── 3.3_generate_sops_for_figure-9(b).R
│   ├── barley_aus_east.rda
│   ├── generate_sop_var.R
│   ├── grids.csv
│   ├── points.csv
│   ├── sopvars_new.rda
│   └── sopvars.rda
├── 04_figures and tables/
│   ├── 0_figure-5_distribution_sops.R
│   ├── 1.1_figure-6(a).R
│   ├── 1.1_indiv_original_results.csv
│   ├── 1.2_figure-6(b).R
│   ├── 1.2_indiv_sop_results.csv
│   ├── 2.1.1_2sopv_pair_results.csv
│   ├── 2.1.1_figure-7(a).R
│   ├── 2.1.2_v_sopv_pair_results.csv
│   ├── 2.1.2-figure-7(c).R
│   ├── 2.2.1_sop_categ_pairs_results.csv
│   ├── 2.2.1-figure-7(b).R
│   ├── 2.2.2_v_sop_categ_results.csv
│   ├── 2.2.2-figure-7(d).R
│   ├── 2.2.3_v_categ_results_individual.csv
│   ├── 2.2.4_vsop_categ_results_individual.csv
│   ├── 3.1_difference_individual_v_v+sop.csv
│   ├── 3.1_figure-8(a)(b)_difference_individual_v_v+sop.R
│   └── 3.2_figure-8(c)(d)_difference_category_c_c+sop.R

```

1. Original environmental & soil data (GEE)

Environmental and soil variables were extracted using Google Earth Engine and aggregated to the SA2/grid level.

- NDVI — MODIS MOD13A2
- NPP — MODIS MOD17A3HGF
- ETa — TERN Landsat AET
- Temperature — ERA5-Land
- Precipitation — ERA5-Land
- VPD — derived from ERA5-Land
- Soil properties — CSIRO SLGA:
 - SOC
 - NTO
 - PTO
 - pHc
 - AWC
- Elevation — Geoscience Australia DEM

MODIS NDVI

MODIS NPP

ERA5-Land

https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_DAILY_AGGR

TERN AET

https://developers.google.com/earth-engine/datasets/catalog/TERN_AET_CMASET_LANDSAT_V2_2

CSIRO SLGA Soil

https://developers.google.com/earth-engine/datasets/catalog/CSIRO_SLGA

DEM (Geoscience Australia)

https://developers.google.com/earth-engine/datasets/catalog/AU_GA_DEM_1SEC_v10_DEM-S

Core GEE scripts

Script	Purpose
1.1_gee_env_grid_2023_2024.js	Extract environmental variables to grids
1.2_gee_soil_extract.js	Extract soil properties to grids

2. Barley production data (ABS)

Folder: 02_original barley_from ABS/

Dataset

Australian Bureau of Statistics — Broadacre Crops
Indicator: Barley production normalised by SA2 area (ton/km²)
Spatial unit: SA2
Years: 2023–2024

Source

ABS Agriculture Statistics:

<https://www.abs.gov.au/statistics/industry/agriculture/australian-agriculture-broadacre-crops/latest-release>

Files

File	Description
2.1_ABS_Original_SA2_Barley_2023_2024.csv	Raw ABS export
2.2_SA2_Barley_2023_24.csv	Cleaned table
2.3_data_aggregated.R	Aggregation script
SA2_2016_AUST.shp	SA2 boundary file used for spatial matching

3. Core SOP & PD analysis code

Folder: 03_core codes/

These scripts implement:

- SOP generation
- Spatial pattern construction
- GeoDetector-based PD computation
- SOP–SOP interaction testing
- Original variable–SOP interaction testing
- Buffer-distance sensitivity analysis

Core scripts

Script	Function
3.1_soh_generate_sops.R	Generate spatial outlier patterns (SOPs)
3.2_calculate_pd.R	Compute PD values using GeoDetector
3.3_generate_sops_for_figure-9(b).R	SOP generation for buffer experiments
generate_sop_var.R	SOP variable generation

Core data objects

File	Description
barley_australia_east.rda	Main working dataset
grids.csv	Grid-level variables
points.csv	Sample barley points
sopvars.rda	SOP variables
sopvars_new.rda	Updated SOP set for buffer experiments

4. Figures & tables — script mapping

Folder: 04_figures and tables/

Each figure in the paper has a corresponding R script.

Figure mapping

Figure	Script
Figure 5	0_figure-5_distribution_sops.R
Figure 6(a)	1.1_figure-6(a).R
Figure 6(b)	1.2_figure-6(b).R

Figure	Script
Figure 7(a)	2.1.1_figure-7(a).R
Figure 7(b)	2.2.1-figure-7(b).R
Figure 7(c)	2.1.2-figure-7(c).R
Figure 7(d)	2.2.2-figure-7(d).R
Figure 8(a,b)	3.1_figure-8(a)(b)_difference_individual_v_v+sop.R
Figure 8(c,d)	3.2_figure-8(c)(d)_difference_category_c_c+sop.R
Figure 9(a)	4.1_figure-9(a)_overall_pd_bar.R
Figure 9(b)	4.2_figure-9(b)_pd_across_buffers.R

Tables (CSV outputs)

Folder: 04_figures and tables/

These CSV files store intermediate and final PD (Power of Determinant) results derived from GeoDetector analyses. Each table corresponds to a specific experimental design and is directly linked to one or more figures in the paper.

Table CSV	Description	Used in Figure
1.1_indiv_original_results.csv	PD values for each individual original variable	Fig. 6(a)
1.2_indiv_sop_results.csv	PD values for each individual SOP variable	Fig. 6(b)
2.1.1_2sopv_pair_results.csv	PD results for SOP–SOP variable pairs	Fig. 7(a)
2.1.2_v_sopv_pair_results.csv	PD results for original variable + SOP variable pairs	Fig. 7(c)
2.2.1_sop_categ_pairs_results.csv	PD results for SOP category pair combinations	Fig. 7(b)
2.2.2_v_sop_categ_results.csv	PD results for variable + SOP category combinations	Fig. 7(d)
2.2.3_v_categ_results_individual.csv	PD values for individual variable categories	Fig. 8(c)
2.2.4_vsop_categ_results_individual.csv	PD values for individual variable+SOP categories	Fig. 8(d)
3.1_difference_individual_v_v+sop.csv	PD difference: variable vs variable+SOP	Fig. 8(a,b)
3.2_v_c_c+sop_individual.csv	PD difference: category vs category+SOP	Fig. 8(c,d)

5. Shapefiles (GIS visualisation)

Folder: 05_shapefiles/

These shapefiles are used for:

- Spatial mapping
- SOP visualisation
- Study region masking
- ArcGIS map production

Core spatial layers

Shapefile	Purpose
SA2_2016_AUST.shp	National SA2 boundaries
SA2_barley_east_bound_pro.shp	Study area boundary
SA2_barley_east_poly_pro.shp	Study area SA2 polygons
SA2_Barley_pro.shp	Barley SA2 units
SA2_East_grid.shp	Analysis grid
STE_2021_AUST_simp_pro.shp	State boundaries

All final maps in the paper were produced using ArcGIS based on these shapefiles.

Requirements & Usage

Language and platforms

- R
- Google Earth Engine
- ArcGIS

Main analytical components

- Generation of spatial outlier patterns (SOPs)
- GeoDetector-based PD calculation
- Individual variable and SOP effect assessment
- SOP–SOP interaction analysis
- Original variable–SOP interaction analysis
- Buffer-distance sensitivity analysis
- Figure and table reproduction

Key parameters for SOP generation

- pointlocation: Sample coordinates
- gridlocation: Grid coordinates
- gridvar: Variables of the grid cells
- distuf: Buffer distance for spatial calculation

- sd: Standard deviation threshold for identifying local outliers

Notes

- SOH = Second-Dimension Outlier-Driven Heterogeneity
- SDO = Second-Dimension Outlier
- SOP = Spatial Outlier Pattern
- PD = Power of Determinant
- All PD values are computed using the GeoDetector framework
- All grid-level variables are aggregated before analysis
- Scripts are numbered in execution order or figure-generation order
- All spatial datasets should remain under the GDA_1994_Australia_Albers coordinate system to maintain consistency

Data and Code Availability

The data and code supporting this study are publicly available through both Figshare and GitHub.

- Figshare: <https://doi.org/10.6084/m9.figshare.31169701>
- GitHub: https://github.com/renkaigis/Second-dimension_Outlier-driven_Heterogeneity

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Contact

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